



M-Tech Production & Marketing

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Week 4 - Day 3 Report

Internship Program — Batch 2026

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1. Introduction

Experiment No. 15 focused on LangChain and AI Agents, introducing me to a powerful framework designed to simplify the development of applications powered by Large Language Models (LLMs). LangChain provides developers with the tools to build complex AI workflows by connecting language models with external data sources, tools, and memory systems. One of the most exciting capabilities enabled by LangChain is the creation of AI Agents — intelligent systems that can reason about tasks, decide which actions to take, and iteratively work toward solving problems using available tools. This experiment was crucial for understanding how LLM applications are built beyond simple prompt-response interactions and how autonomous agents can be developed to handle complex, multi-step tasks.

2. Objectives

- Understand the purpose and capabilities of the LangChain framework.
 - Learn how chains combine prompts, language models, and outputs into reusable workflows.
 - Explore the concept of AI agents and their decision-making processes.
 - Understand the think-act-observe loop that agents use to solve problems.
 - Study how agents can be equipped with tools such as calculators, search engines, and APIs.
 - Learn how memory enables conversational agents to maintain context across multiple interactions.
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3. Task Description

Experiment No. 15 involved studying LangChain's core components and their role in building LLM applications. The concept of chains was explored, including how prompts and language models can be connected into reusable pipelines. The experiment then focused on AI agents, examining how they differ from simple chains by incorporating decision-making capabilities and tool usage.

The agent workflow was studied in detail, including how agents receive a task, decide which tools to use, execute actions, observe results, and continue iterating until they reach a satisfactory solution. The role of memory in maintaining context across conversational turns was also examined to understand how agents can engage in meaningful multi-turn interactions.

4. Work Done

Experiment No. 15 began with studying the LangChain framework and its role in connecting language models with external systems. The concept of PromptTemplate was explored, understanding how reusable prompts can be created with placeholders for dynamic content, then connected to language models using the chain operator.

Different types of tools available for agents were examined, including calculators for mathematical operations, search tools for retrieving information, and custom tools for specialized tasks. The agent initialization process was studied, understanding how agents are configured with a language model, a set of tools, and a reasoning strategy such as the zero-shot ReAct approach.

The think-act-observe loop was explored in detail, demonstrating how agents repeatedly analyze tasks, select appropriate actions, execute those actions using available tools, observe the results, and decide on next steps until they can provide a complete answer to the user's query.

Memory functionality was also studied, understanding how conversational memory allows agents to recall previous interactions and use that context to provide more coherent and relevant responses in ongoing conversations.

5. Challenges Faced

One of the main challenges in Experiment No. 15 was understanding how agents decide which tools to use for different tasks and how they interpret tool outputs to inform their next actions. This required careful analysis of the reasoning process and how observations from tool usage feed back into the agent's decision-making.

Another challenge involved understanding the distinction between simple chains and agents. While chains follow predetermined execution paths, agents dynamically choose their own paths based on task requirements and intermediate results. This conceptual shift from linear pipelines to autonomous decision-making required focused study of agent architecture and reasoning mechanisms.

6. Key Learnings

Experiment No. 15 provided valuable knowledge of the LangChain framework and its capabilities for developing LLM applications. I learned how chains can streamline repetitive tasks by connecting prompts and models into efficient workflows. I also gained a deeper understanding of AI agents and how they use reasoning, observation, and action to autonomously solve complex problems.

The concept of the think-act-observe loop was particularly insightful, demonstrating how agents iteratively refine their approach based on intermediate results. I also learned about the importance of memory in building conversational applications and how agents can maintain context across multiple interactions.

Additionally, I gained practical insight into how LangChain enables developers to build sophisticated AI applications such as research assistants, customer service bots, and automated task execution systems by combining language models with external tools and data sources.

7. Conclusion

Experiment No. 15 successfully demonstrated the power and flexibility of modern frameworks for building LLM applications. By understanding how chains, agents, tools, and memory work together, developers can create intelligent systems capable of handling complex, multi-step tasks that go far beyond simple question-answering.

This experiment expanded my understanding of how language models can be integrated into broader application architectures, equipped with tools to interact with external systems, and empowered to make autonomous decisions. The concepts learned provide a strong foundation for developing advanced AI applications that can reason, plan, and execute tasks in real-world scenarios.

Intern Signature:



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Designation: